MLT Week 6

Goodness of Maximum Likelihood Estimator for Linear Regression

We observed that $w^* = \hat{w}_{ML} = (XX^T)^+ Xy$ when noise follows zero mean Gaussian distribution.

Now we want a way to understand how good $\hat{w}_{ML}$ is in estimating w.

$$= E\left[\|\hat{w}_{ML} - w\|^2\right] \quad \begin{pmatrix}\text{Expected value of the norm of} \\ \text{estimator \& } w \in \mathbb{R}^d\end{pmatrix} \text{ [over randomness in } y\text{]}$$

$$= \sigma^2 \cdot \text{trace}\left[(XX^T)^+\right]$$

- The goodness of the estimator is not just by the noise but by how the features are related to each other.
- We cannot touch σ^2 as it is given but we can fool around with the features to get a better estimator with lesser mean squared error.

Cross-validation for minimizing MSE

$\text{trace}\left[(XX^T)^{-1}\right]$ [sum of diagonal entries] also [sum of eigenvalues of matrix]

Let the eigenvalues of XX^T be $\{\lambda_1, \dots, \lambda_d\}$ ∴ Eigenvalues of $(XX^T)^{-1}$: $\left\{\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_d}\right\}$

Mean Squared Error$(\hat{w}_{ML}) = E\left[\|\hat{w}_{ML} - \hat{w}\|^2\right] = \sigma^2 \sum_{i=1}^{d} \frac{1}{\lambda_i}$

For some matrix A, let eigenvalues be $\{\lambda_1, \dots, \lambda_d\}$

Then what are the eigenvalues of $(A+\lambda I)$? $= \{(\lambda_1+\lambda), \dots, (\lambda_d+\lambda)\}$ ↳ $\lambda \in \mathbb{R}_{+ve}$, $I \in \mathbb{R}^{d \times d}$

$$\text{trace}\left((XX^T + \lambda I)^{-1}\right) = \left(\sum_{i=1}^{d} \frac{1}{\lambda_i + \lambda}\right)$$

We want to decrease MSE by reducing the value of trace. How do we find this $\lambda \in \mathbb{R}_+$?

By the method of **cross validation**

training set | validation set

training set (80%)	validation set (20%)

- Train on training set & check for errors on validation set
- Pick λ which gives least error

How to decide the best way to divide the data?

K-fold cross validation

$$[F_1|F_2|\cdots|F_K]$$

- Train on folds $= \{f_1, \ldots f_{i-1}, f_{i+1}, \ldots, f_k\}$
- Validate on f_i
- Pick λ which gives least avg error

$$\boxed{\hat{w}_{new} = (XX^T + \lambda I)^{-1} Xy}$$

Bayesian Modeling for linear regression

Need a prior on w, i.e., $P(w) \in \mathbb{R}^d$

Likelihood: $y/x \sim N(w^T x, 1)$ ↳ for simplicity, can use σ^2 as well

A choice of Prior: $w \sim N(0, \gamma^2 I)$ $\begin{bmatrix} \gamma^2 & & \\ & \ddots & \\ & & \gamma^2 \end{bmatrix}$ covariance matrix

We know,

$$P[w/\{(x_1, y_1), \ldots, (x_n, y_n)\}] \propto P[(x_1, y_1), \ldots, (x_n, y_n)/w] \cdot P(w)$$

$$\propto \left[\prod_{i=1}^{n} e^{-\frac{(y_i - w^T x_i)^2}{2}}\right] \left[\prod_{i=1}^{d} e^{-\frac{(w_i - 0)^2}{2\gamma^2}}\right]$$

$$\propto \left[\prod_{i=1}^{n} e^{-\frac{(y_i - w^T x_i)^2}{2}}\right] \left[e^{\frac{1}{2\gamma^2}\left(-\sum_{i=1}^{d} w_i^2\right)}\right] = \left[e^{-\frac{\|w\|^2}{2\gamma^2}}\right]$$

$$\hat{w}_{MAP} = \max_w \sum_{i=1}^{n} -\frac{(y_i - w^T x_i)^2}{2} - \frac{\|w\|^2}{2\gamma^2}$$

$$= \left(\min_w \sum_{i=1}^{n} \frac{(y_i - w^T x_i)^2}{2} + \frac{\|w\|^2}{2\gamma^2}\right) f(w)$$

Take gradient, set it to 0 & solve for $\hat{w}_{MAP}$

$$\nabla f(w) = (XX^T)w - Xy + \frac{w}{\gamma^2}$$

$$\hat{w}_{MAP} = \left[XX^T + \frac{I}{\gamma^2}\right]^{-1} Xy$$

Ridge Regression

$$\hat{w}_R = \underset{w}{\text{argmin}} \underbrace{\sum_{i=1}^{n}(w^T x_i - y_i)^2}_{\text{Loss}} + \underbrace{\lambda\|w\|^2}_{\text{Regularizer}}$$

↳ Ridge